# Superposition and interference

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Add signed displacements and distinguish interference from lost energy.

**SUCCESS CRITERIA - I can...**

1. Add wave displacements at the same place and time.
2. Recognize constructive and destructive interference.
3. Explain that pulses continue after crossing in a linear medium.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Across a stationary boundary, frequency remains 3 Hz while speed changes from 15 to 21 m/s. Find both wavelengths.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $y_{resultant} = y_1 + y_2$.
Add signed displacements at the same place and time. Local cancellation does not destroy all wave energy.

**Worked reasoning:** At one point, two pulses have displacements +3 cm and -1 cm. Superposition gives +2 cm there. Add signed displacements, not their absolute values.

**Your turn:** Draw the two component displacements and the resultant at that point.

A

B

Sum

Add signed displacements; label units.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Two upward pulses overlap. Sketch the combined displacement. Then sketch equal upward/downward pulses overlapping.



Sketch and label your response.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Two coherent sound waves have equal amplitude A at a microphone. Draw their contributions and resultant when phase difference is 0°, then 180°. Compare resultant amplitudes. What must stay stable for the pattern to persist?

A

B

Sum

Add signed displacements; label units.

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

Two rope pulses momentarily cancel. A student says the pulses are gone forever. What happens after they pass in an ideal linear medium?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A process has $\Delta U$=-24 J and $W_{by}$=32 J. Find Q and decide whether heat enters or leaves.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3

## 07 Exit check

Three instantaneous displacements are +5,-2,-6 mm. Find resultant and state whether this single value determines all wave energy.

A

B

Sum

Add signed displacements; label units.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

Two waves at a point have equal amplitudes 4 cm. Find maximum constructive amplitude and complete destructive amplitude; explain the required relative phases.

A

B

Sum

Add signed displacements; label units.

**Marzano check - circle:** 1 2 3